

```
In [2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
import matplotlib.font_manager as fm

# ===== 1. 设置字体 =====
font_path = 'C:/Windows/Fonts/msyh.ttc' # 字体路径
fm.FontManager.addfont(font_path)
prop = fm.FontProperties(fname=font_path)
font_name = prop.get_name()

plt.rcParams['font.family'] = font_name
plt.rcParams['axes.unicode_minus'] = False

sns.set_style("whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)

# ===== 2. 加载数据 =====
df = pd.read_csv('digital_manufacturing_dataset.csv')
print("数据形状: ", df.shape)
print("\n前5行: ")
display(df.head())
print("\n数据类型: ")
print(df.dtypes)
print("\n统计摘要: ")
display(df.describe())

# ===== 3. 数据清洗 =====
# 3.1 重复值
initial_rows = len(df)
df = df.drop_duplicates()
print(f"删除重复行数: {initial_rows - len(df)}")

# 3.2 缺失值
print("\n缺失值统计: ")
print(df.isnull().sum())

# 3.3 数值列缺失值用中位数填充
for col in df.select_dtypes(include=['number']).columns:
    if df[col].isnull().any():
        median_val = df[col].median()
        df[col].fillna(median_val, inplace=True)
        print(f"{col} 列缺失值已用中位数 {median_val} 填充")

# 3.4 异常值检测 (生产速率)
Q1 = df['Production_Rate'].quantile(0.25)
Q3 = df['Production_Rate'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
outliers = df[(df['Production_Rate'] < lower) | (df['Production_Rate'] > upper)]
print(f"Production_Rate 异常值数量: {len(outliers)}")

# ===== 4. 探索性分析 =====
# 4.1 目标变量分布
plt.figure()
```

```

sns.histplot(df['Production_Rate'], bins=30, kde=True)
plt.title('生产速率分布', fontproperties=prop)
plt.xlabel('生产速率', fontproperties=prop)
plt.ylabel('频次', fontproperties=prop)
# 刻度标签字体
for label in plt.gca().get_xticklabels():
    label.set_fontproperties(prop)
for label in plt.gca().get_yticklabels():
    label.set_fontproperties(prop)
plt.show()

# 4.2 分组统计
group_stats = df.groupby('Demand_Class').agg({
    'Production_Rate': ['mean', 'std', 'count'],
    'Machine_Utilization': 'mean',
    'Downtime_Minutes': 'mean'
}).round(2)
print(group_stats)

# ===== 5. 可视化对比（三张柱状图） =====
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# 5.1 生产速率
sns.barplot(x='Demand_Class', y='Production_Rate', data=df, ax=axes[0])
axes[0].set_title('平均生产速率', fontproperties=prop)
axes[0].set_xlabel('需求等级', fontproperties=prop)
axes[0].set_ylabel('生产速率', fontproperties=prop)
for label in axes[0].get_xticklabels():
    label.set_fontproperties(prop)
for label in axes[0].get_yticklabels():
    label.set_fontproperties(prop)
means = df.groupby('Demand_Class')['Production_Rate'].mean()
for i, v in enumerate(means):
    axes[0].text(i, v + 0.5, f'{v:.1f}', ha='center', fontproperties=prop)

# 5.2 机器利用率
sns.barplot(x='Demand_Class', y='Machine_Utilization', data=df, ax=axes[1])
axes[1].set_title('平均机器利用率', fontproperties=prop)
axes[1].set_xlabel('需求等级', fontproperties=prop)
axes[1].set_ylabel('利用率 (%)', fontproperties=prop)
for label in axes[1].get_xticklabels():
    label.set_fontproperties(prop)
for label in axes[1].get_yticklabels():
    label.set_fontproperties(prop)
means2 = df.groupby('Demand_Class')['Machine_Utilization'].mean()
for i, v in enumerate(means2):
    axes[1].text(i, v + 0.5, f'{v:.1f}', ha='center', fontproperties=prop)

# 5.3 停机时间
sns.barplot(x='Demand_Class', y='Downtime_Minutes', data=df, ax=axes[2])
axes[2].set_title('平均停机时间（分钟）', fontproperties=prop)
axes[2].set_xlabel('需求等级', fontproperties=prop)
axes[2].set_ylabel('停机时间（分钟）', fontproperties=prop)
for label in axes[2].get_xticklabels():
    label.set_fontproperties(prop)
for label in axes[2].get_yticklabels():
    label.set_fontproperties(prop)
means3 = df.groupby('Demand_Class')['Downtime_Minutes'].mean()
for i, v in enumerate(means3):
    axes[2].text(i, v + 0.5, f'{v:.1f}', ha='center', fontproperties=prop)

```

```

plt.tight_layout()
plt.savefig('three_metrics_comparison.png', dpi=300)
plt.show()

# ===== 6. 相关性热力图 =====
plt.figure(figsize=(12, 10))
corr = df.corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)
plt.title('特征相关性矩阵', fontproperties=prop)
plt.savefig('correlation_heatmap.png', dpi=300)
plt.show()

# 与生产速率最相关的特征
top_features = corr['Production_Rate'].abs().sort_values(ascending=False).head(6)
print("与生产速率最相关的特征：")
print(top_features)

# ===== 7. 统计检验 (ANOVA) =====
group0 = df[df['Demand_Class'] == 0]['Production_Rate']
group1 = df[df['Demand_Class'] == 1]['Production_Rate']
group2 = df[df['Demand_Class'] == 2]['Production_Rate']

f_stat, p_value = f_oneway(group0, group1, group2)
print(f"ANOVA 结果: F统计量 = {f_stat:.3f}, p值 = {p_value:.4f}")

if p_value < 0.05:
    print("结论: 不同需求等级之间的生产速率存在显著差异 (p < 0.05)")
else:
    print("结论: 不同需求等级之间的生产速率无显著差异")

```

数据形状: (9800, 17)

前5行:

	Machine_Temperature	Process_Temperature	Rotational_Speed	Torque	Tool_W
0	72.483571	159.338104	1147.103930	49.872934	
1	69.308678	155.032324	1091.239681	38.827144	
2	73.238443	173.056368	1168.011570	36.796821	
3	77.615149	139.844186	1089.863831	24.386342	
4	68.829233	158.225729	1173.960893	34.879098	

数据类型:

Machine_Temperature	float64
Process_Temperature	float64
Rotational_Speed	float64
Torque	float64
Tool_Wear	int64
Vibration_Level	float64
Production_Rate	float64
Machine_Utilization	float64
Downtime_Minutes	int64
Inventory_Level	int64
Supplier_Lead_Time	int64
Logistics_Cost	float64
Order_Quantity	int64
Sales_Value	float64
Customer_Demand_Index	float64
Transaction_Frequency	int64
Demand_Class	int64

dtype: object

统计摘要:

	Machine_Temperature	Process_Temperature	Rotational_Speed	Torque
count	9800.000000	9800.000000	9800.000000	9800.000000
mean	69.989632	150.128705	1197.750143	39.890988
std	5.022667	10.001145	197.930235	8.050275
min	50.387999	113.647998	428.724934	4.275169
25%	66.642300	143.360582	1059.809297	34.410353
50%	69.987025	150.140861	1199.455612	39.881477
75%	73.358173	156.885502	1332.408254	45.400921
max	89.631189	194.790843	1938.324900	69.822667

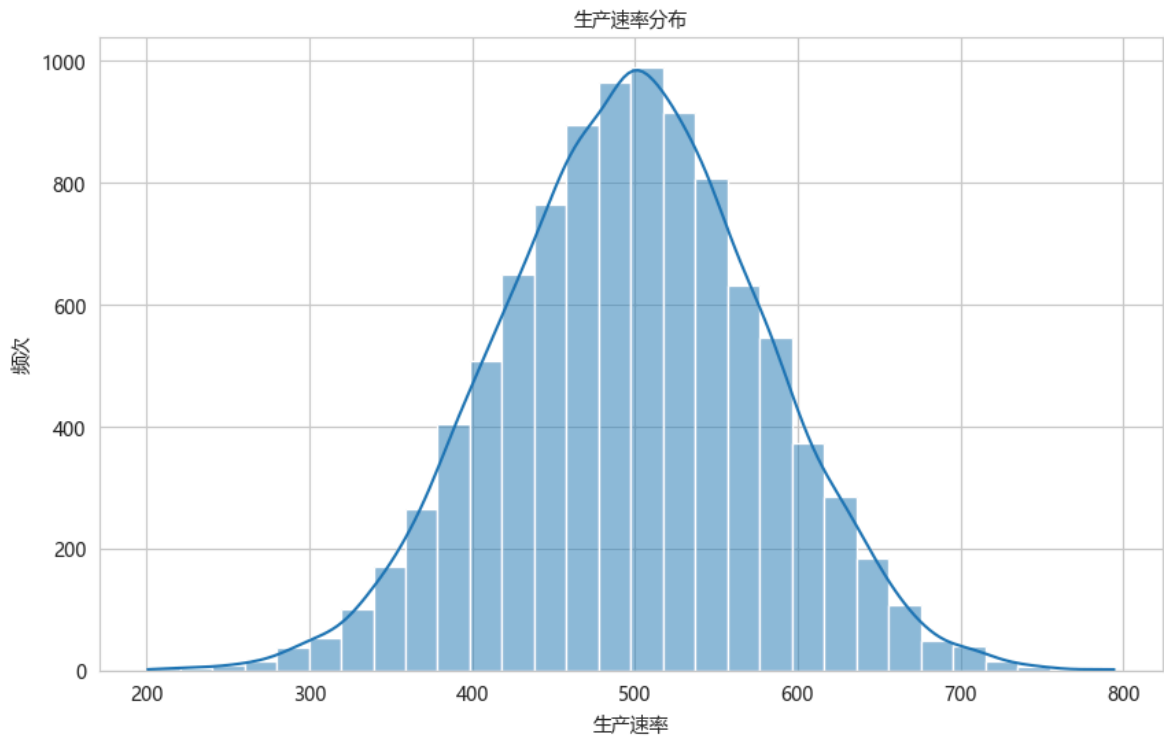
删除重复行数: 0

缺失值统计:

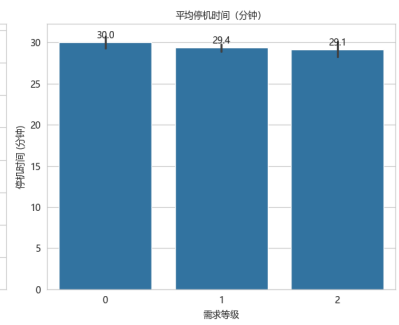
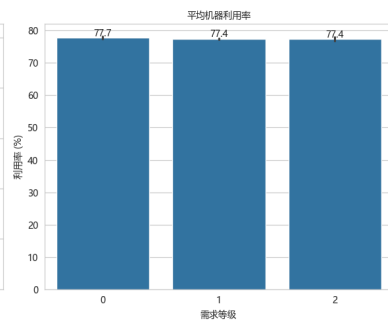
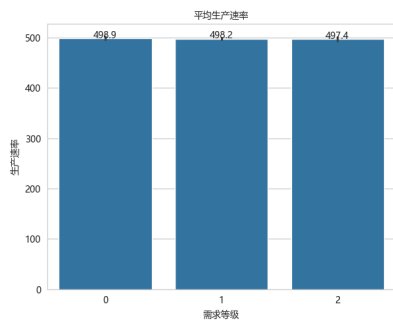
Machine_Temperature	0
Process_Temperature	0
Rotational_Speed	0
Torque	0
Tool_Wear	0
Vibration_Level	0
Production_Rate	0
Machine_Utilization	0
Downtime_Minutes	0
Inventory_Level	0
Supplier_Lead_Time	0
Logistics_Cost	0
Order_Quantity	0
Sales_Value	0
Customer_Demand_Index	0
Transaction_Frequency	0
Demand_Class	0

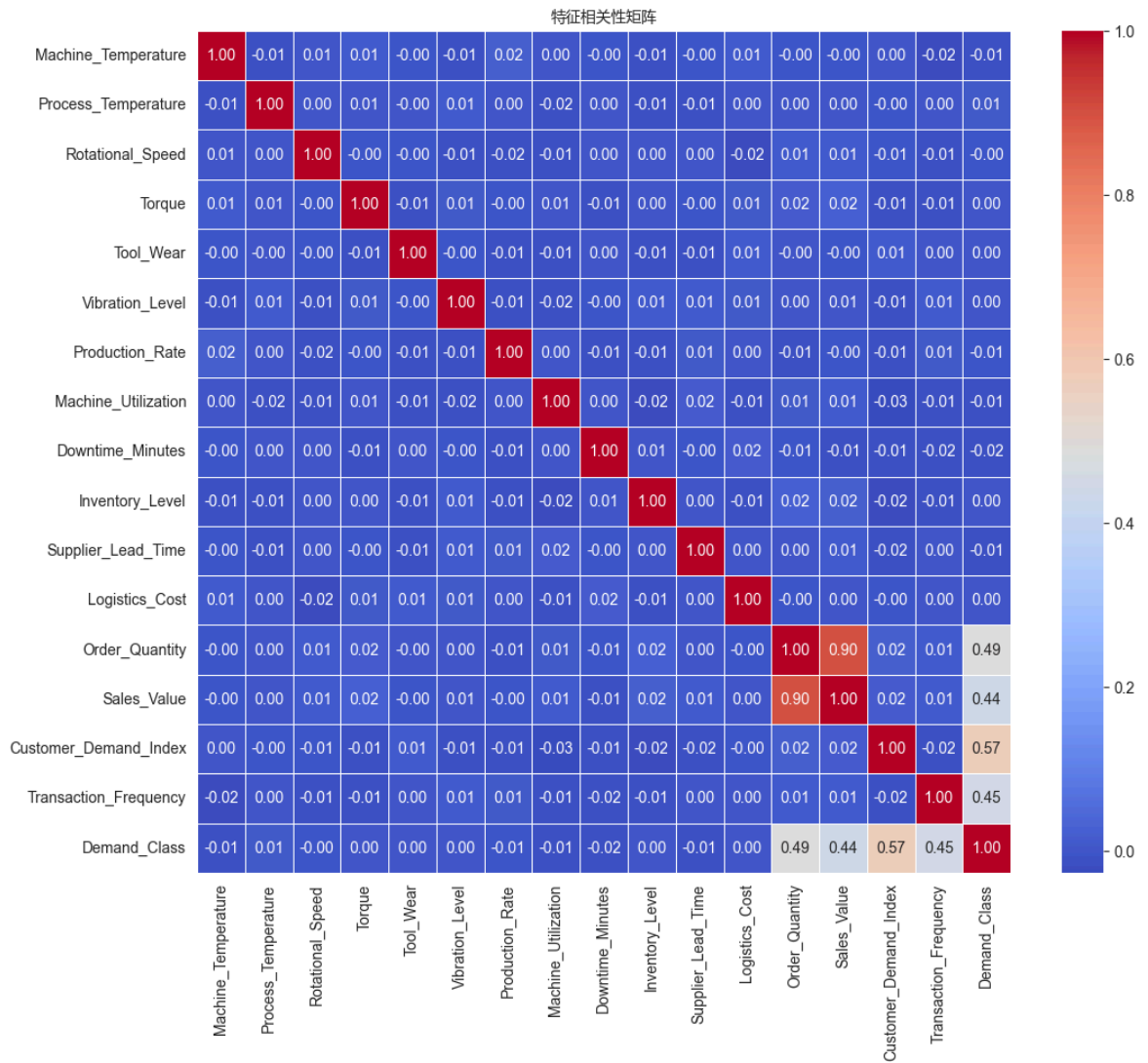
dtype: int64

Production_Rate 异常值数量: 69



Demand_Class	Production_Rate			Machine_Utilization		Downtime_Minutes
	mean	std	count	mean		mean
0	498.92	79.75	2291	77.72		30.01
1	498.16	79.98	6172	77.45		29.37
2	497.41	78.73	1337	77.41		29.12





与生产速率最相关的特征：

Production_Rate 1.000000
Machine_Temperature 0.019113
Rotational_Speed 0.015767
Customer_Demand_Index 0.014910
Tool_Wear 0.014610
Vibration_Level 0.014257

Name: Production_Rate, dtype: float64

ANOVA 结果：F统计量 = 0.159, p值 = 0.8529

结论：不同需求等级之间的生产速率无显著差异